

## Class Meeting Handout #27

Barry Linkletter

Hydrolysis of esters in acid or base is the classic example of an acyl transfer reaction. But there is much more. Acyl transfer is how amide bonds are made in ribosomes. We will explore some basics and then enjoy the example of solid-phase synthesis of peptides.<sup>1</sup>

This document was produced using the L<sup>A</sup>T<sub>E</sub>X typesetting language with the Tufte-handout document class. Chemical diagrams were created in ChemDoodle and modified using Affinity Designer.

<sup>1</sup> Read Chapter 10.17 of the textbook

### Acyl Transfer Reactions

*Discussion: Acyl Groups – 10 minutes*

Acyl groups are seen everywhere in biological chemistry and so are a popular subject of physical organic chemistry investigations.

1. Identify the acyl groups below and describe where you would find them in biochemistry? If one is not found in living systems, propose a analogous acyl group that is.

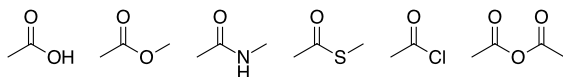


Figure 1: Common acyl groups.

←

2. Comment on the reactivity of the acyl groups above for transfer to water (hydrolysis).

*Exercise: Mechanisms of Ester Hydrolysis – 15 minutes*

3. Propose the standard mechanism for ester hydrolysis of ethyl acetate in acidic and basic water.
4. Hydrolysis of t-Butyl acetate does not proceed via either of the mechanisms proposed above. It is unable to be hydrolyzed in basic water but is very rapidly hydrolyzed in acidic water. Propose a mechanism. Propose an experiment that would differentiate this mechanism from the usual mechanism for acid-catalyzed hydrolysis.

*Exercise: Making Amides – 10 minutes*

Emil Fischer was awarded the second Nobel prize in chemistry in 1903 in recognition of his total synthesis of glucose. Following that prize, he turned to the synthesis of peptides in an effort to reveal the structure of proteins.<sup>2</sup>

5. Figure 2 on the next page presents his synthesis of a dipeptide of glycine.
  - (a) This reaction would give a very low yield of product. Explain why.

<sup>2</sup> "The bold legacy of Emil Fischer." M. Goodman, W. Cai, N.D. Smith, *J. Peptide Sci.*, 2003, 9, 594–603. <https://doi.org/10.1002/psc.476>

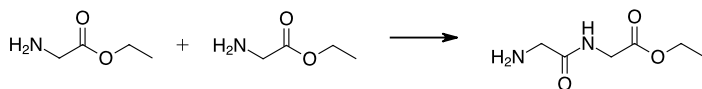


Figure 2: Fishers early method for peptide synthesis.

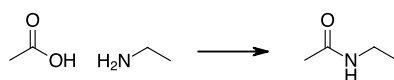
- (b) What conditions would be required to ensure the maximum yield (still a poor yield to be sure, but you'll get nothing unless you choose your conditions wisely)
- (c) How could be improve the chemistry?

*Discussion: Solid-Phase Peptide Synthesis – 15 minutes*

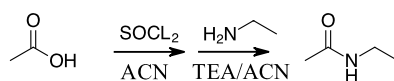
There are better ways to make peptides.<sup>3</sup>

6. Let us make a peptide linkage using a few options. Consider the reactions below.

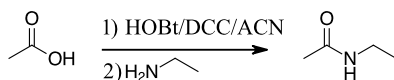
- (a) Would the following reaction produce an amide bond?



- (b) What is the mechanism for this amide synthesis. How does it compare to the proposed reaction above? Are there any downsides to this reaction for making oligopeptides?



- (c) Have you seen this one before? What is the mechanism? What is the driving force of this reaction?



7. Consider the methods of solid-phase peptide synthesis as presented in Figure 3 on the following page.

- (a) What chemical features should we look for in the protecting groups and the linker group? How do the protecting groups used here demonstrate these features?
- (b) How does this rate on the Green Chemistry Scale?<sup>4</sup>

<sup>3</sup> You may already have been introduced to this chemistry in previous courses. You can review the idea of peptide synthesis and the solid-phase method in Chapter 26.7 and 26.8 of McMurry. <https://openstax.org/details/books/organic-chemistry>. It is also examined in Chapter 10.17 of our textbook. Although I abhor websites as references, the Wikipedia article for solid-phase peptide synthesis is also a great place to start. [https://en.wikipedia.org/wiki/Peptide\\_synthesis](https://en.wikipedia.org/wiki/Peptide_synthesis)

<sup>4</sup> I know its bad, but people are trying to do better. "Greening Fmoc/tBu solid-phase peptide synthesis." O. Al Musaimi, B.G. de la Torre, F. Albericio, *Green Chem.*, 2020, 22, 996–1018. <http://dx.doi.org/10.1039/C9GC03982A>

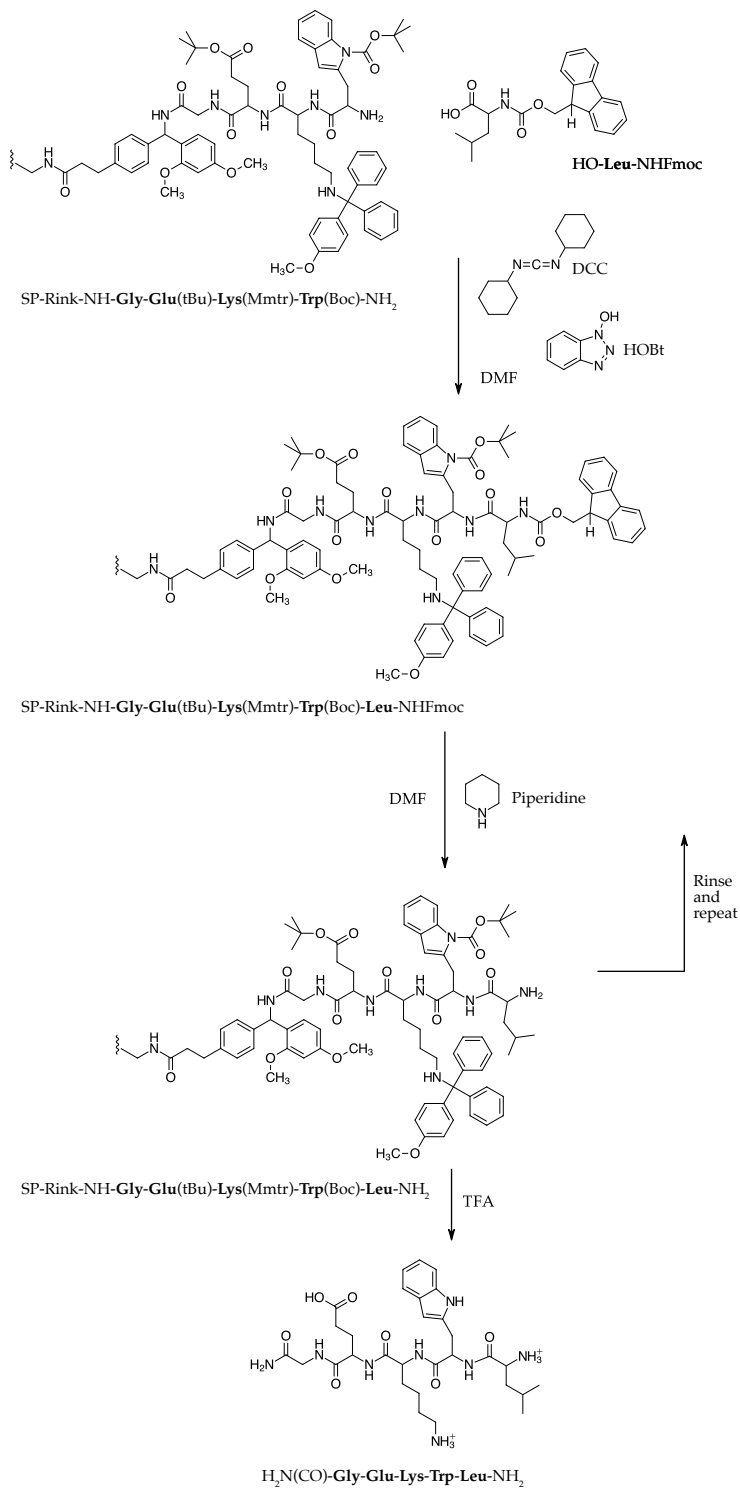


Figure 3: An example of solid-phase synthesis of peptides in action.

Consider these fun activities...

a) Identify the atoms of the amino acids in the peptide and in the leucine building-block being added.

b) Identify the new peptide bond. Circle and identify all the protecting groups

c) What is the mechanism for the removal of the Fmoc group?

d) What is the mechanism for the cleavage of everything (protecting groups and linker group)?

e) the Rink-amide linker results in a carboxamide group at the C-terminus. Will this be a problem for you? What would you do if you wanted a natural carboxylic acid at the C-terminus?